

Wilderness First Aid

Instructor Whiteboard Guide



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Each page shows the recommended whiteboard layout for that module.

Module 1: Provider Safety

Scene Assessment & Consent

SCENE SIZE-UP

1. STOP — Pause, assess from distance
2. LOOK — Identify all hazards
3. THINK — Safe? PPE? More help?

IF UNSAFE → Do NOT enter. Call for help.

Personal Protective Equipment

- Nitrile gloves (2 pairs)
 - Eye protection
 - Mask / face shield
 - Hand hygiene AFTER removal
- Gloves ON before EVERY contact

Consent Types

- EXPRESSED: Alert, understands, agrees
- IMPLIED: Unconscious or minor
- REFUSAL: Document, witness, do not abandon

Scope of Practice

- Assess → Treat life threats →
- Stabilize → Evacuate
- Do not exceed your training

A dead responder helps no one. Safety first — always.

Module 2: Patient Assessment

XABCDE Primary + SAMPLE/OPQRST

X	eXsanguination: STOP life-threatening hemorrhage
A	Airway: Open? Clear? Maintain?
B	Breathing: Rate / Effort / Effectiveness
C	Circulation: Pulse, skin signs, shock?
D	Disability: AVPU + pupils + neuro
E	Expose/Environment: Find injuries, prevent hypothermia

A = ALERT: Oriented ×4

V = VERBAL: Responds to voice

P = PAIN: Responds to pain only

U = UNRESPONSIVE: No response

SAMPLE

S — Signs & Symptoms
A — Allergies

OPQRST

O — Onset
P — Provocation/Palliation

Every patient, every time: XABCDE first, then SAMPLE. Document everything.

E — Events leading up

T — Time/Trend

Module 3: Airway Management

Position & Techniques

Position by Condition

Unconscious, breathing → Recovery position

Respiratory distress → Sit up, lean forward

Shock (no spinal) → Supine, legs elevated 6-12"

Spinal concern → Neutral supine — do NOT move

Head-Tilt Chin-Lift

No spinal injury suspected

One hand on forehead

Fingers under chin — lift

Tilt head back gently

Jaw Thrust

Spinal injury suspected

Fingers behind jaw angles

Push jaw forward

No head tilt

OBSTRUCTION

Conscious: Encourage cough, back blows

Abdominal thrusts

Unconscious: Open, look, listen, feel

Attempt ventilation

Open airway → Rescue breath if apneic. No blind finger sweeps.

Module 4: Bleeding & Wound Management

Control & Dressing

STOP THE BLEED — In Order

1. DON PPE — Gloves NOW
 2. DIRECT PRESSURE — Firm, 10+ minutes
 3. If soaks through — ADD more on top
 4. If extremity & uncontrolled — TOURNIQUET
- Apply 2-3" above wound, write time

ARTERIAL

Bright red, spurting
Life-threatening

VENOUS

Dark red, steady
Serious

CAPILLARY

Oozing
Minor

Special Wounds

IMPALED: Stabilize in place, do NOT remove

OPEN CHEST: Occlusive dressing, 3-sided seal

EVISCERATION: Moist dressing, do NOT push in

AMPUTATION: Direct pressure + keep cool/moist

Direct pressure 10 minutes — don't peek. Tourniquet = life over limb.

Module 5: Shock

Recognition & Treatment

EARLY SIGNS | LATE SIGNS

Anxiety, restlessness
Tachycardia (HR >100)
Pale, cool, clammy skin
Capillary refill >2 sec

Altered LOC
Very rapid, weak pulse
Mottled/cyanotic skin
Severe BP drop

When LOC drops → critical. Treat early.

Field Treatment — In Order

1. Control hemorrhage (X in XABCDE)
2. Open/maintain airway
3. POSITION: Supine, legs elevated 6-12"
4. WARMTH: Prevent hypothermia
5. ORAL FLUIDS: If alert, not nauseated
6. EVACUATE — patient needs care

HYPOVOLEMIC

Blood/fluid loss → control, position, fluids

ANAPHYLACTIC

Severe allergy → Epi NOW

NEUROGENIC

Spinal injury → NO leg elevation

SEPTIC

Infection → antibiotics, evacuate

Treat early. Warm. Position. Fluids. Evacuate. Shock kills quietly.

Module 6: Musculoskeletal Injuries

CSM & Splinting

CSM Assessment — Before AND After Every Splint

C — CIRCULATION: Pulse? Capillary refill <2 sec?

S — SENSATION: Can feel light touch distal?

M — MOVEMENT: Can wiggle fingers/toes?

Abnormal CSM = URGENT — realign or evacuate

The 6 Rules of Splinting

1. Assess CSM BEFORE splinting

2. Immobilize joint ABOVE and BELOW fracture

3. PAD bony prominences

4. Splint in position of COMFORT

5. Secure firm, NOT tight

6. Reassess CSM AFTER — every 30 minutes

Upper Extremity

Clavicle: Sling + swathe

Shoulder dislocation: Sling comfort

Humerus/Forearm/Wrist: Splint + sling

Lower Extremity

Femur: Traction splint if available

1-2L blood loss! URGENT

Knee/Tibia/Fibula: Straight splint

Ankle: Pillow splint or SAM

Open Fractures

Cover wound (do NOT push bone back)

Irrigate if able

Splint as found

Antibiotics if available

URGENT evacuation

CSM before and after every splint. Treat femur fractures as shock.

Module 7: Spine & Head Injuries

Clearance & TBI

Can You Clear the Spine?

- Alert and oriented × 4
 - NO midline tenderness
 - NO neurological deficits
 - NO distracting injuries
 - NOT intoxicated
- If ANY = NO → Immobilize & Evacuate

Concussion Signs

- Headache, dizziness
- Light/sound sensitivity
- Memory gaps
- Nausea
- Any concussion = NO return to activity

EVACUATE URGENTLY If:

- Altered or decreasing LOC
- Worsening or severe headache
- Confusion / combativeness
- Vomiting more than once
- Unequal pupils
- Skull deformity or Battle's sign
- Weakness one side of body

In-Line Stabilization

- Hands both sides of head
- Do NOT apply traction
- Maintain NEUTRAL position
- Logroll: count '1-2-3 roll'
- Maintain alignment

When in doubt, immobilize. A missed spinal injury is catastrophic.

Module 8: Heat Emergencies

Cramps → Exhaustion → Stroke

HEAT CRAMPS

Signs:

Muscle cramps, sweating

Treatment:

Rest, fluids + electrolytes, stretch

HEAT EXHAUSTION

Signs:

Sweating, weakness, nausea

Treatment:

Shade, cool, fluids, monitor

HEAT STROKE

Signs:

ALTERED LOC, hot skin

Treatment:

COOL FIRST, then transport

Rapid Cooling — Heat Stroke

1. Cold water immersion (MOST effective)
 2. Ice packs to neck, armpits, groin
 3. Cool wet sheets + aggressive fanning
 4. Shade + remove all insulation
- COOL FIRST. Don't wait for transport.

Critical Difference

EXHAUSTION: Normal/near normal LOC

STROKE: ALTERED LOC

Treat stroke as EMERGENCY

LOC change = heat stroke. Cool aggressively. Every minute matters.

Module 9: Cold Emergencies

Hypothermia & Frostbite

Hypothermia Severity

MILD (95-90°F): Shivering, confusion, SHIVERING IS GOOD

MODERATE (90-86°F): Shivering STOPS (danger!), confusion, rigidity

SEVERE (<86°F): Unconscious, may appear dead, rigid

Handle gently. 'Not dead until warm and dead'

Field Rewarming — In Order

1. Stop further heat loss FIRST
2. Move from cold/wet environment
3. Remove wet clothing — dry replacement
4. Insulate from ground
5. Vapor barrier layer
6. Heat packs: armpits/groin — NOT direct skin
7. Warm sweet liquids: mild only, alert patient
8. Gentle handling — NO rough movement

Frostbite Recognition

SUPERFICIAL: Waxy, pale, firm

but NOT frozen — treat now

DEEP: Hard frozen, white/gray/black,

do NOT rewarm if re-freeze risk

Frostbite Treatment

Rewarm in 104°F water ONLY if no re-freeze risk

Protect tissue — do NOT rub, no dry heat

Large blisters may form — leave intact

Hypothermic until warm and dead. Handle gently — cold hearts are irritable.

Module 10: Anaphylaxis

Recognition & EpiPen

EARLY Signs

Hives
Flushing
Itching
Anxiety
Runny nose

SEVERE Signs

Throat tightening
Wheezing
Face/tongue swelling
Hypotension

Systemic reaction after trigger = ANAPHYLAXIS
Can kill in minutes — act immediately

Epinephrine — Auto-Injector

1. Remove blue safety cap
2. Press firmly into OUTER THIGH
3. Can inject THROUGH clothing
4. Hold 10 seconds
5. Then massage site
6. May repeat in 5-15 min

DOSING:

Adult (>30kg): 0.3mg

Child (<30kg): 0.15mg

Full Anaphylaxis Protocol

1. Epinephrine FIRST — immediately
 2. Call 911 / initiate evacuation
 3. Position: Hypotensive → legs up
 4. Antihistamines (secondary)
 5. O2 if available
 6. Monitor for biphasic reaction (4-12h later)
 7. EVACUATE — epi is temporary
- Antihistamines are NOT sufficient alone

Epi first, always. Antihistamines are NOT a substitute. Evacuate every time.

Module 11: Medical Emergencies

AEIOU-TIPS & Key Conditions

AEIOU-TIPS — Why Is This Patient Altered?

A — Alcohol / Toxins
E — Epilepsy / Seizure
I — Insulin (Diabetic)
O — Overdose / Poisoning
U — Uremia (Kidney)
T — Trauma / Temperature
I — Infection / Sepsis
P — Psychiatric
S — Stroke / Shock

Seizure Management

During: Protect from injury, clear area, time it

After: Recovery position, assess, SAMPLE

Evacuate if: First ever, >5 min, repeated

Diabetic Emergency

LOW (Hypo): Fast, sweating, shaky, confused

→ Give sugar if conscious

HIGH (Hyper): Slow, fruity breath, thirsty

→ Evacuate, cannot fix in field

When unsure → give sugar

STROKE — FAST

F — Face drooping (one side)

A — Arm weakness (one side)

S — Speech slurred or absent

T — Time — call for help NOW

Time is brain. Evacuate urgently.

Altered LOC = AEIOU-TIPS. Treat what you can. When in doubt, evacuate.

Module 12: Bites & Stings

Snakes, Insects, Marine

DO | DON'T

- Stay calm — slow venom spread
- Immobilize limb below heart
- Remove rings/watches (swelling!)
- Mark swelling edge + time
- Evacuate — antivenom is only treatment

- ✗ Cut and suck
- ✗ Tourniquet
- ✗ Ice or cold packs
- ✗ Electric shock (myth)
- ✗ Alcohol

Tick Removal

1. Fine-tipped tweezers — grasp near skin
2. Steady, even upward pull
3. Clean site with soap and water
4. Record date and location

DON'T: Vaseline, heat, twisting

Watch for: Bull's-eye rash, fever, joint pain

Bee/Wasp Stings

Single sting: Remove stinger, cold pack, antihistamine

Multiple stings (>10-20): Monitor, evacuate

Anaphylaxis → Module 10 protocol

Stinger: scrape don't squeeze

Animal Bites

Irrigate thoroughly — most important

Do NOT close the wound

High-risk rabies: bats, raccoons, foxes

All bites → medical evaluation

Marine Injuries

JELLYFISH: Vinegar (Pacific), hot water (NE US)

STINGRAY: Hot water immersion (113°F), remove barb

SEA URCHIN: Hot water soak, do NOT crush spines

Snakebite: immobilize + evacuate — antivenom only treatment. No cut-and-suck.

Module 13: Evacuation Principles

Decision & Methods

What happens if I wait 24 hours?

URGENT (1-2 hours)

Airway compromise
Uncontrolled hemorrhage
Altered LOC
Spinal injury w/ deficits
Heat stroke
Severe hypothermia
Anaphylaxis
Open fracture

NON-URGENT (24 hours)

Stable long bone fracture
Controlled wound
Mild hypothermia (resolved)
Isolated concussion, resolved

Getting Them Out

SELF-RESCUE: Walk with assistance
ASSISTED CARRY: Improvised litter, 4+ carriers
TECHNICAL RESCUE: Ropes, litters
Requires specialized training
HELICOPTER: When above impossible

Helicopter LZ Setup

100' × 100' clear area
Firm, level (<8° slope)
Mark corners with signals
Signal wind direction
Approach from FRONT — NEVER rear

MIST — Handoff to EMS

M — MECHANISM: How did patient get injured?
I — INJURIES found: Describe findings
S — SIGNS & vitals: Current status
T — TREATMENT given: What did you do?
Practice this out loud before EMS arrives

Decide early. Evacuate early. Never delay when in doubt. MIST before handoff.